

How Wrong Is the Computer's Guess?

A plain-language summary of a method that proves a prediction is wrong by at least a certain amount — what it achieves, where it falls short, and how it compares with the other approaches people use.

The short version

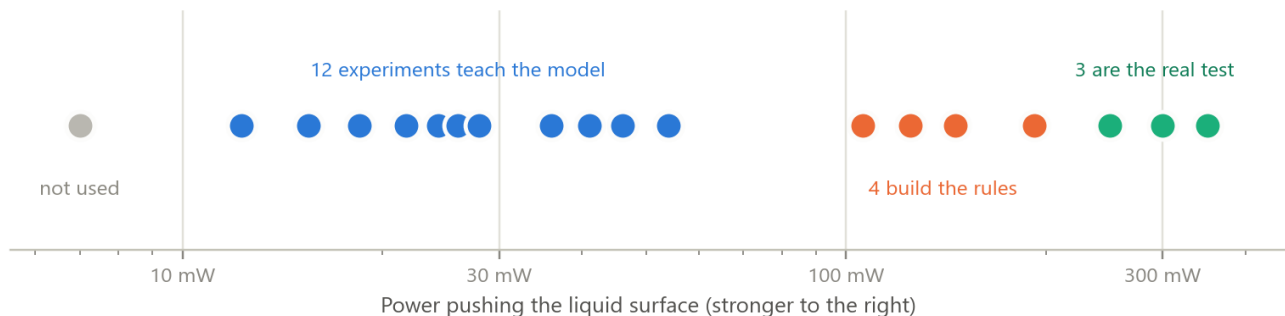
Scientists shook a liquid surface harder and harder and recorded the tiny waves that appeared. A computer model learned from the gentle shakes and was then asked to predict what the strong shakes would look like. The usual question is “was the prediction right?” This project asks a different and much harder one:

Without looking at the answer, can we prove the prediction is wrong — and by at least how much?

The answer is yes, but only if you are willing to assume something about the physics. The method now accounts for roughly **78%** of the straight-line model's real mistake and about **81%** of the small neural network's, using nothing but earlier measurements. The catch is that the whole thing rests on an assumption that has only been checked on three or four experiments.

What was actually measured

Twenty recordings were made of a liquid surface driven at different powers, from a gentle 7 milliwatts to a strong 350. Each recording gets turned into a “fingerprint”: twelve numbers describing how much the surface wobbles slowly versus quickly. The model's job is to predict that fingerprint for a power it has never seen.



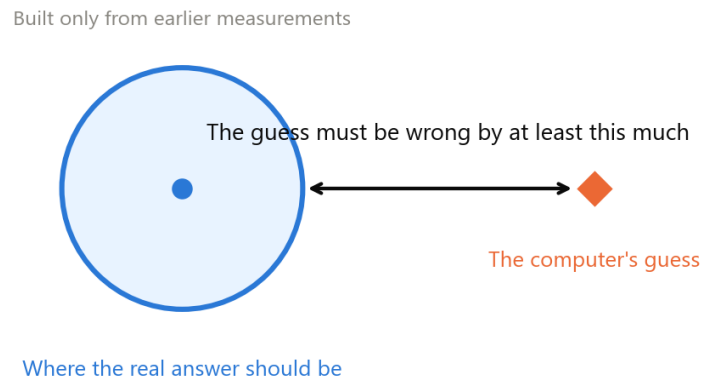
The twenty experiments and what each is used for. Note how far the three test experiments sit from everything the model learned on — that gap is the whole difficulty.

One number to keep in mind

Everything below is measured as a **share of the real mistake we can prove**. If the model's prediction was actually off by 10 units and our method proves it was off by at least 8, that is 80%. It is *not* a measure of how accurate the model is — a high score means our warning is informative, not that the prediction is good. 0% means we cannot prove anything, which is not the same as the prediction being right.

How the method works

Imagine a friend tells you a hidden target is somewhere inside a circle drawn on a map. You do not know exactly where the target is, but you know the circle. Now someone guesses a spot far outside that circle. Even if the target sits at the very nearest edge, the guess is still wrong by the gap between the guess and the circle. You can state a minimum error without ever being told the true location.



The entire method in one picture. The circle comes from measurements we already had; the diamond is the prediction being judged.

Where the circle comes from

The four experiments at 106, 125, 146 and 192 milliwatts were already measured, so their answers are known. Physics suggests the fingerprint should drift in a predictable direction as you turn the power up. Starting from the strongest measured experiment and drifting along that expected direction gives the centre of the circle. How much the drift rule might be off gives its radius. Push further past what was measured and the circle grows, which is exactly right: the further you extrapolate, the less you should claim.

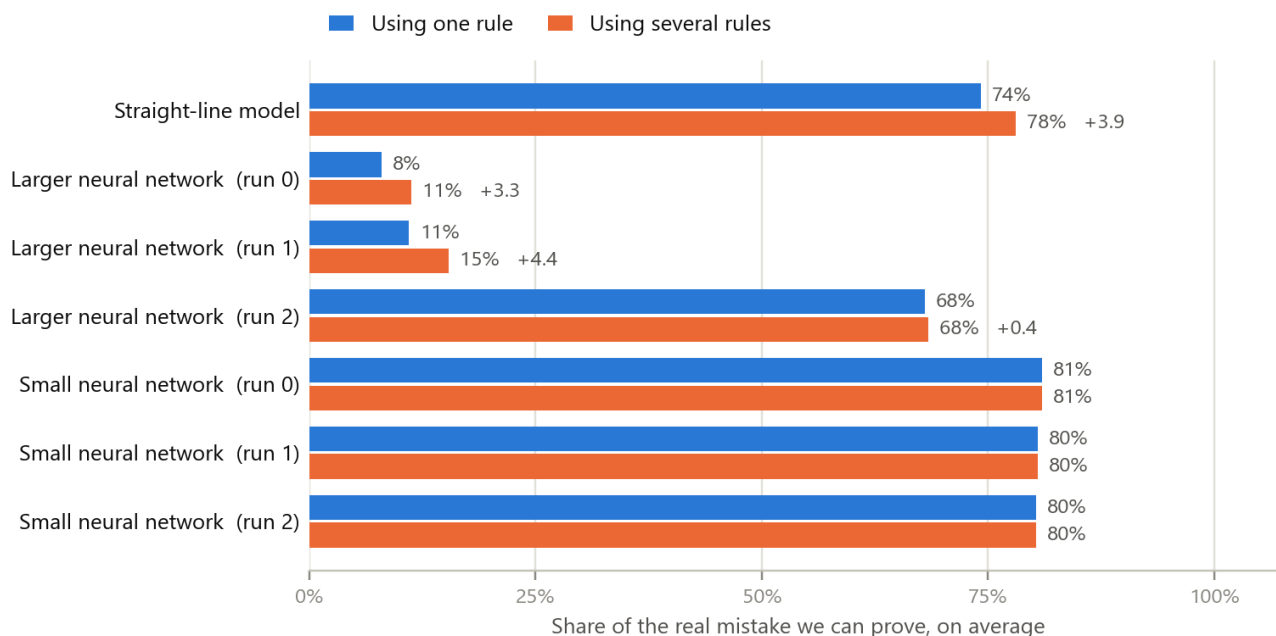
Crucially, the true answer for the test experiment is never used to compute the warning. It is unsealed afterwards, only to check whether it really did land inside the circle.

Why bother, if you could just measure it?

Because in real use there is nothing to measure. A model like this exists so that someone can skip the experiment — predict the outcome at a setting nobody has run, and act on it. At that moment there is no answer to check against, and the model will happily produce a confident-looking prediction whether or not it has any idea what it is doing.

A warning built only from old measurements is something you can actually have in that situation. It will never tell you a prediction is good — it can only catch predictions that are provably bad. A silent warning means “no proof of error found here”, which is a much weaker statement than “this prediction is fine.”

The results



Each row is one model, at the setting where the checker may use measurements up to 192 milliwatts. Longer bars mean a more informative warning. The orange bars are the newest method, described just below.

The straight-line model is a deliberately poor predictor, so its mistakes are large and easy to prove. The larger neural network is the most accurate of the three, and there the method struggles: its errors are small enough to hide inside the circle, so little or nothing can be proven. That is the honest shape of the result — **the worse the model, the louder the warning.**

The newest experiment: combining several rules

Until recently the checker fitted one drift rule using all four calibration experiments. The new idea: fit a separate rule on every group of at least three of them, then report whichever gives the strongest provable error. Because the choice uses only the prediction and the earlier measurements, it is still fair — no peeking at the answer.

It worked, modestly. For the straight-line model the provable share rose from **74.2%** to **78.1%**. For one run of the larger neural network it rose from **11.0%** to **15.4%**. The small neural network barely moved.

Three reasons not to celebrate too hard

- **The number cannot go down.** The original single rule is one of the candidates being compared, and we report the best candidate. So this comparison can only show an increase or no change. A gain of zero is the floor of the test, not a failure of the method.
- **The clever part did nothing.** The method includes machinery for blending several circles together. Across all 78 checks, blending beat simply picking the single best circle by at most 0.0008 — against typical mistakes of about 13. Essentially the entire gain comes from picking one group: the three experiments at 125, 146 and 192 milliwatts, dropping the 106 one.

- **One setting could not improve at all.** When the checker may only use measurements up to 146 milliwatts it has three calibration experiments, and there is exactly one way to choose three from three. Its zero change is arithmetic, not evidence.

And the price: a stronger assumption

The old method needed the answer to be inside **one** circle. The new one needs it inside **all five at once** (up to 42 in the widest test). That held every single time we checked — but in the tightest case the answer cleared the circle's edge by only 0.20 units, where the comfortable cases had room of 1.1 to 1.9. Squeezing a better number out also squeezes the assumption holding it up.

A trap worth naming

It is tempting to report “the proven minimum never exceeded the real error, 78 times out of 78” as if that were a successful test. It is not. When the answer is inside the circles, the mathematics *guarantees* the minimum cannot exceed the real error — the code checks this as an internal consistency check, not as evidence. There is really only one thing being tested here: did the answer land inside the circles? It did, on three to four distinct experiments. That is the whole evidence base.

How this compares with other methods

Almost every well-known technique answers the opposite question. They say “the true answer is probably within this range” — an *upper* limit on how wrong you might be. This project produces a *lower* limit: proof that you are definitely wrong by at least some amount. Very few methods do that, and there is a good mathematical reason why.

Approach	What it tells you	What it needs	How it compares here
Conformal prediction (the standard tool)	A range the answer probably falls in, with a stated success rate	Typically 100-1000 measured examples that resemble the test case	Far stronger guarantees, but needs hundreds of examples. This project has 3-4, so the standard route is simply unavailable
Ensembles and Bayesian neural networks	A spread of opinions across many models, read as confidence	Lots of training and retraining; no measured answers needed	Cheap and popular, but known to be overconfident exactly where this project operates: far outside the training range
Physics-residual methods (e.g. calibrated PINN uncertainty)	How badly the prediction violates a known equation	The governing equation written down explicitly	Needs no answers at all, which is better. But it scores violation of the equation, not the actual error, and no exact equation is available for these waves
Certified error bounds for physics networks	A rigorous ceiling on the error	The exact equation plus a stability constant	Genuinely rigorous, but an upper limit, and reported as loose in practice. Different question from this one
This project	A floor: proof the prediction is off by at least this much	A handful of measured examples plus a physics-based drift assumption	Works with very little data and answers the rarer question. Pays for it with an assumption that is hard to verify

Why the assumption is not a sloppy shortcut

A 2025 theory paper asked whether you can prove a model is wrong without assuming anything about the underlying problem. The answer is essentially no. Once a family of models is flexible enough to fit far more data than you have, every assumption-free lower bound collapses to the useless statement “the error is at least zero.” Their threshold is sharp: it bites once the models could fit roughly the square of your sample size.

So the physics assumption in this project is not a corner being cut. Some such assumption is *mathematically required* for this kind of claim to say anything at all. The fair criticism is not that an assumption exists — it is that this particular one has been tested on only three or four experiments.

A related idea

One other line of work does derive genuine error floors: when a network is built to assume a symmetry that the real world only partly has, that mismatch forces a minimum error. The spirit is the same — a structural assumption about the world converted into a provable floor. This project differs in getting its structure from measured calibration experiments rather than from a symmetry group.

The honest bottom line

Claim	Status
The arithmetic is correct	Solid. The geometry is a standard inequality, and the awkward cases (contradictory circles, ties, the no-improvement case) are covered by unit tests. The published numbers reproduce exactly from a clean copy.
The improvement is real and fairly earned	Yes. No test answer influences the score or the choice of rule. But it is modest, it cannot come out negative, and it comes from picking a group rather than from the blending machinery.
The warnings were correct on the data	Yes, on every one of the 78 checks. Note that this is 3-4 distinct experiments; the 78 comes from repeating across models, random starts and overlapping splits, which add no new evidence.
The method will work on new experiments	Not established, and not establishable with this data. Three successes out of three support a success-rate floor of only about 37%, and even that figure assumes the experiments are independent, which experiments chosen by turning one dial on one rig are not.

One thing that genuinely improved

In an earlier version of this work, the proven minimum was so far below the real error that the check passed no matter what — it could not have detected a broken assumption. That is no longer true. The proven minimum now comes within about 1.0 unit of the real error in the tightest case, and exceeds 90% of it in 16 of the 78 checks. The check now has teeth.

What would actually settle it

- More high-power experiments. Around 30 independent test cases would be needed before any claim about a success rate is even measurable. The current corpus has three.

- Experiments from a second apparatus. Everything here comes from one rig, and the data had already been examined before these checks were designed.
- A test where the assumption is expected to fail. Every check so far passed, which is encouraging but also means the premise has never been observed breaking. A method that has never been seen to fail has not really been stress-tested.

In one sentence

The method does something unusual and useful — it turns a few old measurements into proof that a prediction is wrong — the newest change made it modestly sharper in a way that is real but partly automatic, and the honest limit is not the mathematics but the three experiments it rests on.

Sources for the comparison

- Gopakumar et al., *Uncertainty Quantification of Surrogate Models using Conformal Prediction* (arXiv:2408.09881).
- Gopakumar et al., *Calibrated Physics-Informed Uncertainty Quantification* (arXiv:2502.04406).
- Müller et al., *Are All Models Wrong? Fundamental Limits in Distribution-Free Empirical Model Falsification* (arXiv:2502.06765, COLT 2025) — the impossibility result.
- Certified machine learning: *A posteriori error estimation for physics-informed neural networks* (arXiv:2203.17055).
- Wang et al., *A General Theory of Correct, Incorrect, and Extrinsic Equivariance* (arXiv:2303.04745) — error floors from symmetry mismatch.
- Wave measurements: UC San Diego deposit, doi.org/10.6075/J0WW7HVJ. Code and full results: github.com/rdgbrandon/ad_model_prediction.